

# Attention-Enhanced Agricultural Image Classification Using EfficientNet-B3 with CBAM and XAI

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## Introduction

### What we have done:

- We implemented a deep learning classification pipeline using the Multi-Source Agricultural Image Dataset for Robust Classification and Semantic Segmentation (AMCD).
- We identified and fixed a critical structural data-loss bug caused by the dataset's original two-level nested hierarchy (Categories  $\rightarrow$  Species).
- We successfully prepared, flattened, and split a comprehensive dataset containing 5,405 images across 77 distinct agricultural classes.

### Novelty:

- We developed an automated structural flattening algorithm utilizing symlinks to elevate nested leaf-node species folders into a unified, top-level class structure.
- Our algorithm features intelligent de-duplication, automatically appending parent category prefixes to classes with identical names (e.g., differentiating between Flowers\_Rose and Vegetables\_Rose).
- This zero-data-loss approach ensures full compatibility with PyTorch's standard ImageFolder loader without requiring physical duplication of files.

## Methods

### Dataset Analysis

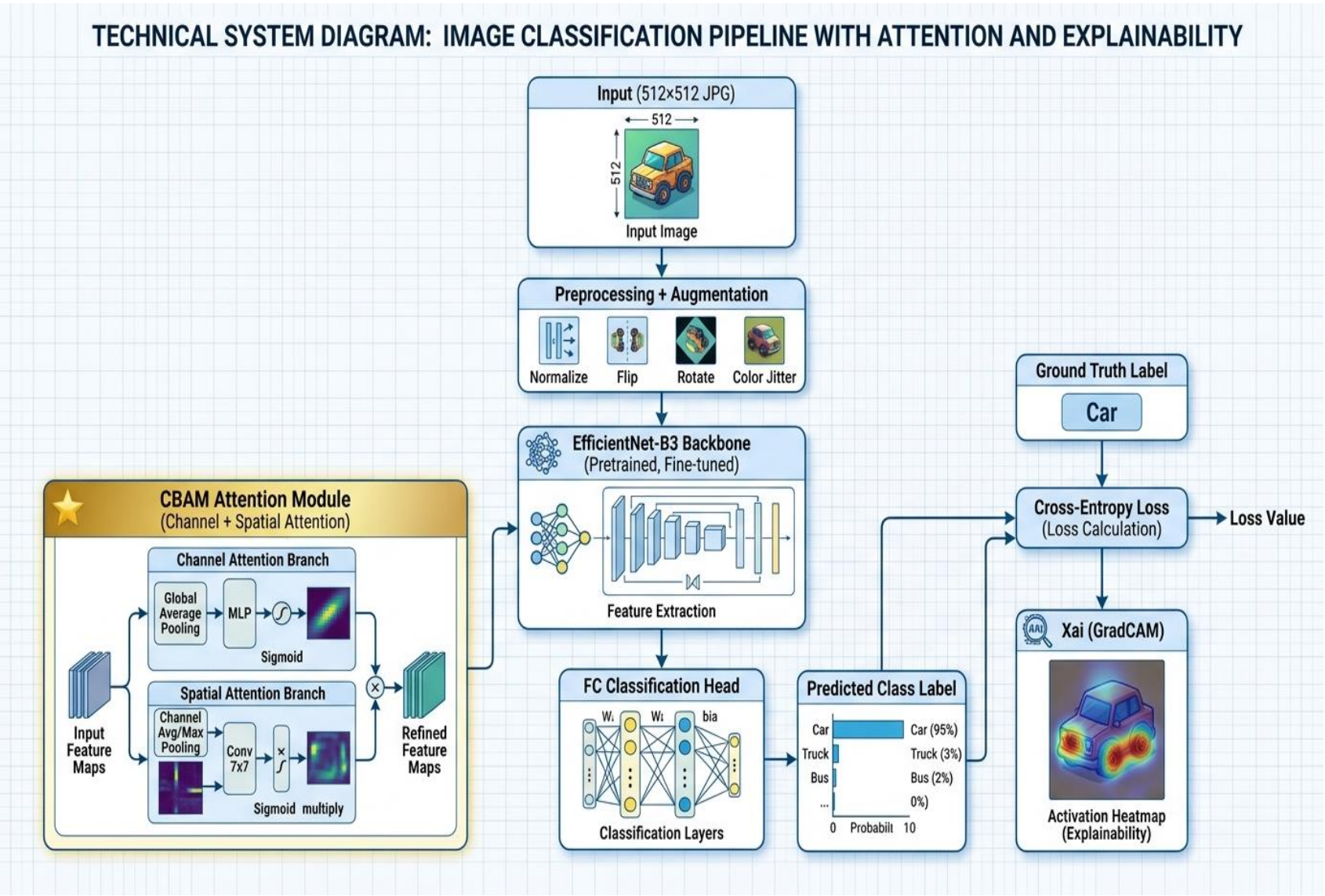
- **Total Images:** 5,405 images.
- **Total Classes:** 77 flattened species/classes.
- **Original Categories Consolidated:** 4 main categories (Crops or Grains, Flowers, Fruits, Vegetables).
- **Dataset Splits:**
  - Training Set: 3,783 images.
  - Validation Set: 810 images.
  - Testing Set: 812 images.

### System Diagram

1. **Raw Input:** Extracted AMCD Directory (4 Categories).
2. **Directory Traversal:** Identify all leaf-node directories containing image files.
3. **Conflict Resolution:** Check for duplicate species name and apply parent-category prefixes.
4. **Symlink Generation:** Map all leaf folders to a single flattened directory (/content/AMCD\_flat).
5. **Data Loading:** PyTorch ImageFolder loads the complet 77-class unified dataset.
6. **Pipeline Integration:** GPU-accelerated (CUDA) batch processing and model augmentation.

### Hyperparameter List

- **Batch Size:** 32.
- **Train Batches:** 119.
- **Validation Batches:** 26.
- **Test Batches:** 26.
- **Hardware / Device:** CUDA (GPU enabled).



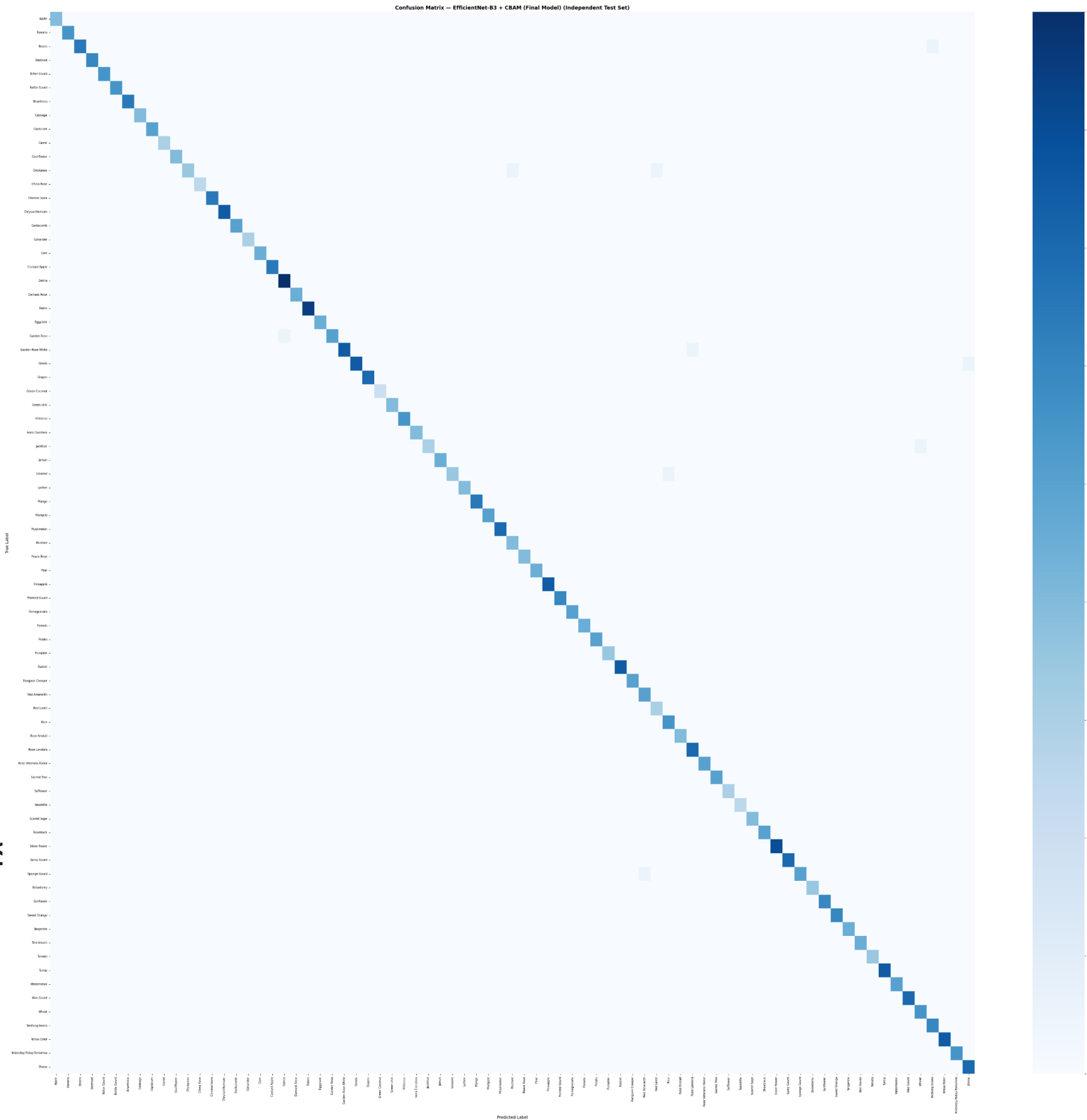
System diagram

## Result

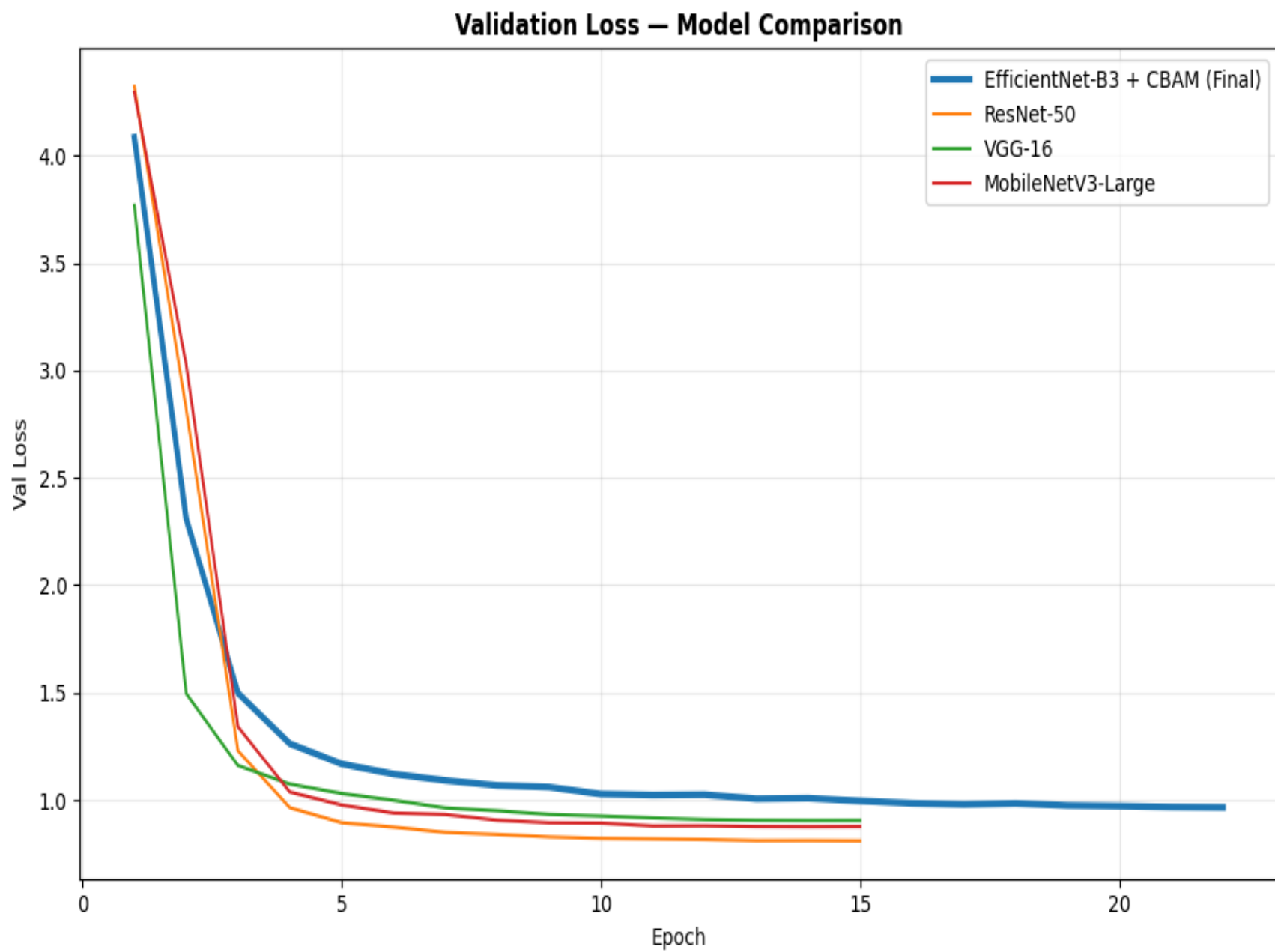
An image-classification pipeline (Google Colab) for an agricultural dataset — **77 classes** (fruits, vegetables, flowers, grains) totaling **5,405 images** (3,783 train / 810 val / 812 test). Main model: **EfficientNet-B3 backbone + CBAM attention + custom FC head**, compared against three baselines (ResNet-50, VGG-16, MobileNetV3-Large) and a CBAM-vs-no-CBAM ablation, plus Grad-CAM visualization.

| Model                         | Top-1 Acc | F1 (macro) |
|-------------------------------|-----------|------------|
| EfficientNet-B3 + CBAM (main) | 98.89%    | 0.9887     |
| ResNet-50                     | 99.51%    | 0.9946     |
| VGG-16                        | 98.52%    | 0.9817     |
| MobileNetV3-Large             | 99.26%    | 0.9907     |

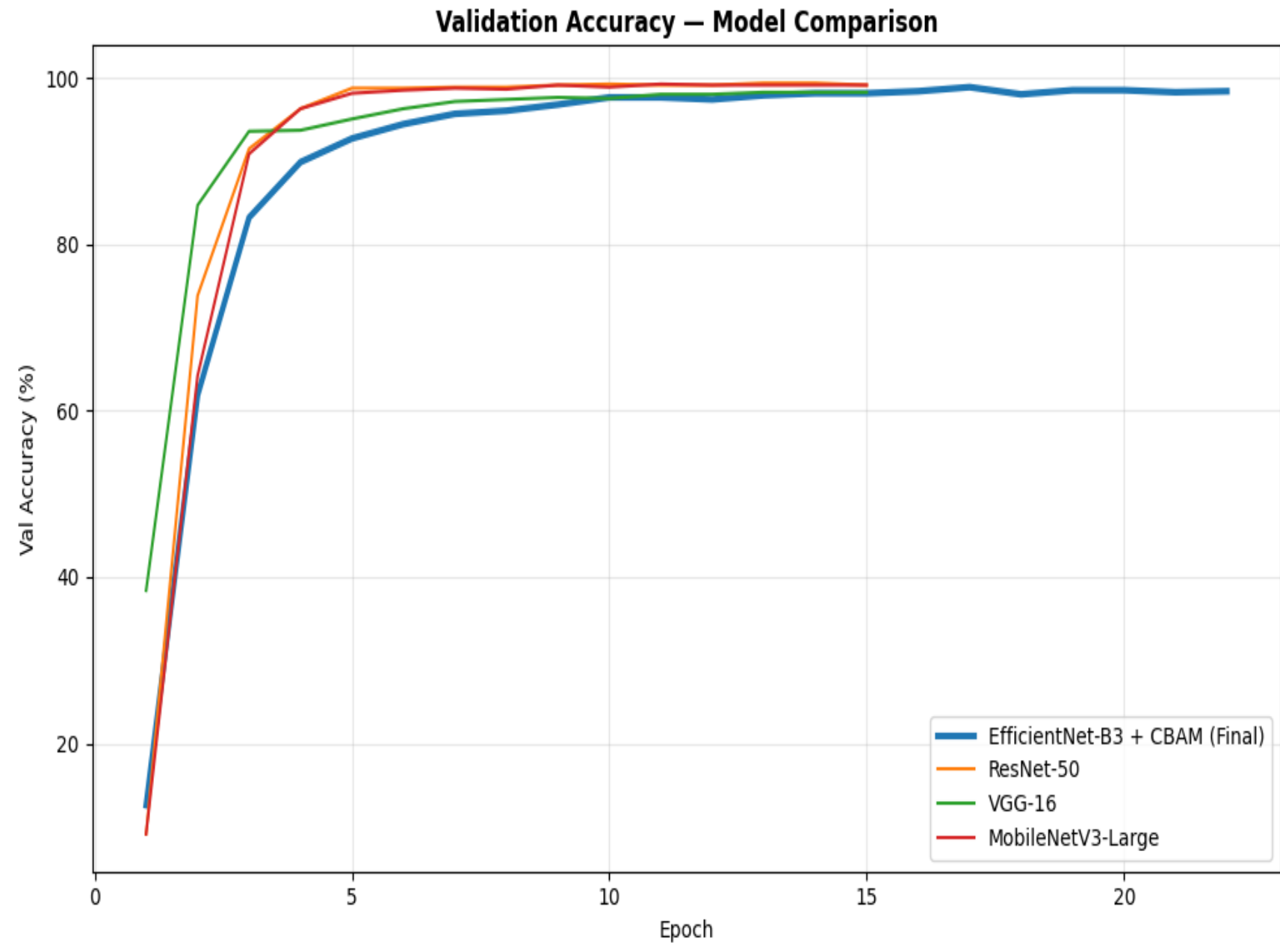
Table: All models top-1 accuracy and F1



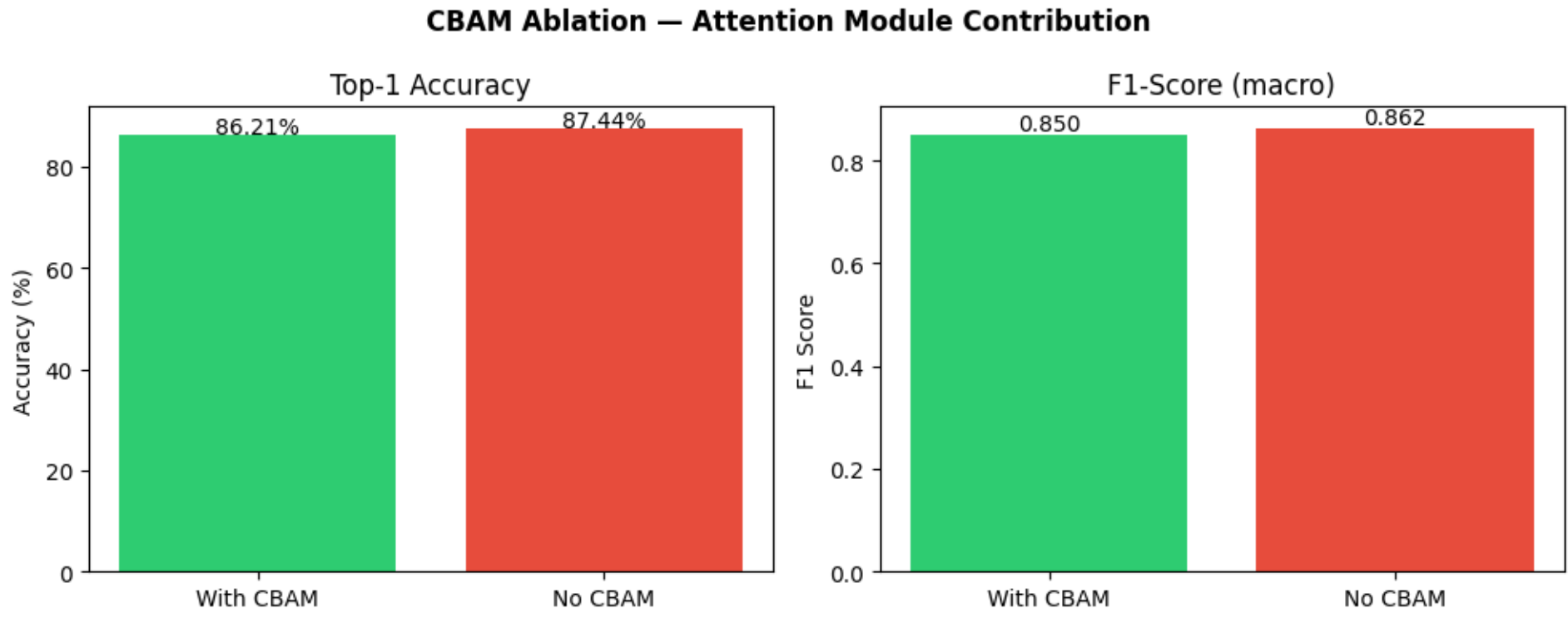
Confusion Matrix: EfficientNet-B3+CBM



Validation loss of all model

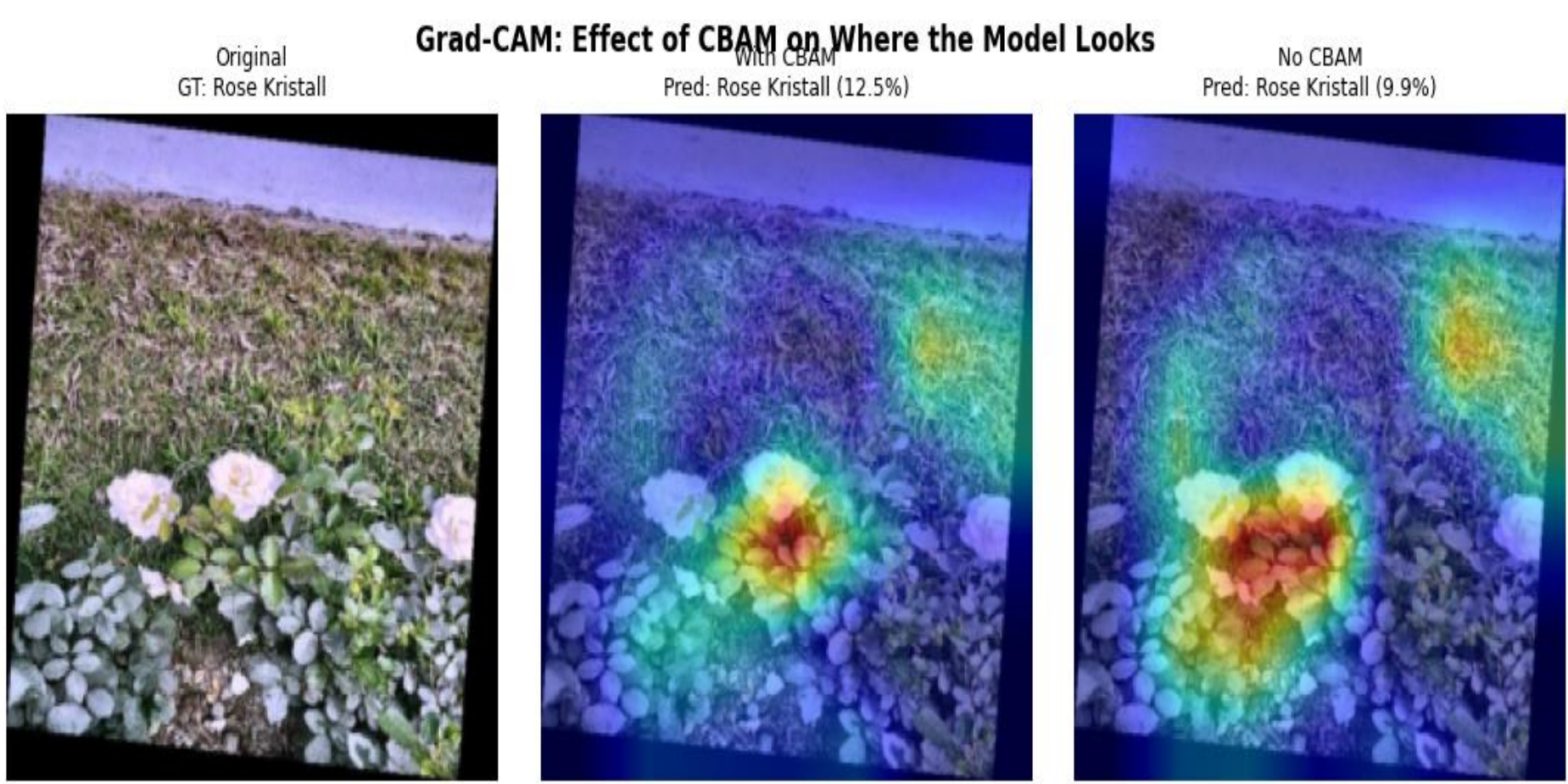


Validation accuracy of all models



CBAM ablation – Attention module contribution

**The headline finding: plain ResNet-50 beats the proposed CBAM-attention model.** The CBAM ablation (Cell 28) confirms this isn't noise — under matched training conditions, removing CBAM *improved* accuracy (86.21% with CBAM  $\rightarrow$  87.44% without, a **-1.23pp** drop from adding CBAM). So across two separate experiments, CBAM attention is not helping this dataset/architecture combo — it's mildly hurting it.



XAI: GradCAM Hit Map

## Conclusion

All four models hit strong accuracy (98.5–99.5%), but the project's core idea — that CBAM attention improves EfficientNet-B3 — **isn't supported**. Plain ResNet-50 beat the CBAM model (99.51% vs. 98.89%), and the controlled ablation confirmed it: removing CBAM *raised* accuracy by 1.23pp. CBAM adds complexity without adding value here, likely because these classes are already visually distinct enough that extra attention isn't needed.

Other notes: the same classes (Chickpeas, Beans, Mustard, Red Lentil, Corn, Tangerine) underperform across every model — a data issue (low sample counts), not a model issue. MobileNetV3-Large is the efficiency pick (99.26% accuracy, lighter than ResNet-50).

**Recommendation:** drop CBAM; use ResNet-50 (best accuracy) or MobileNetV3-Large (best accuracy-per-cost).